

UNCONDITIONAL DOUBLY EXPONENTIAL UPPER BOUNDS FOR THE BOTTOM OF WINDOWED WEIL QUADRATIC FORMS

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ABSTRACT. For $\lambda > 1$ let $\lambda_{\min}(\lambda)$ be the bottom of the spectrum of Weil's quadratic form restricted to test functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$. By Weil's criterion, the Riemann Hypothesis is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ . We prove two *upper* bounds that hold without any hypothesis on the zeros of ζ . Theorem A gives $\lambda_{\min}(\lambda) \leq C\lambda^p e^{-2\pi\lambda^2}$, with all constants explicit; for every $\lambda \geq \sqrt{5}$ one may take $\lambda_{\min}(\lambda) \leq 1.563 \times 10^9 \lambda^{16} e^{-2\pi\lambda^2}$. Theorem B improves the exponent to $e^{-4\pi\lambda^2}$ by means of Kaiser–Bessel windows. Both proofs rest on one observation: the image of the summation map $\varphi \mapsto \sum_n \varphi(nu)$ lies in the radical of the Weil form, and cutting such a function off to the window costs only a Gaussian tail. The exponent 4π is the exponent of the prolate eigenvalue defect $1 - \chi_2(\lambda)$, which Connes, Consani and Moscovici observed to track $\lambda_{\min}$ numerically. We extend this comparison to the certified numerical upper bounds of Zhu: divided by $1 - \chi_2$, they stay between 3 and 20 over 283 orders of magnitude. Theorem A is formalized in Lean 4, on top of a formalization of Weil's explicit formula; the formal statement depends only on the standard axioms. These results concern upper bounds only. They say nothing about positivity and do not address the Riemann Hypothesis.

Status.

- **Theorem A.** Proved in this paper, with explicit constants established by ball arithmetic. In its form with existential constants it is formalized in Lean 4 (declaration `RhWeil.Concrete.theoremA`). That formalization depends only on `propext`, `Classical.choice` and `Quot.sound`, and its statement was audited against the standard form of the explicit formula.
- **Theorem B.** Proved in this paper; not formalized. The constants C and p are explicit; the threshold λ_0 is not effective.

This paper has not yet been reviewed by independent human experts.

1. INTRODUCTION

1.1. Weil's criterion on windows. Weil's criterion states that the Riemann Hypothesis (RH) holds if and only if a certain quadratic form W is nonnegative on test functions [16, 2]. Connes and Consani [3] and Connes, Consani and Moscovici [4] study the restriction W_λ of W to functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$. Write $\lambda_{\min}(\lambda)$ for the bottom of its spectrum. Then:

- $\lambda_{\min}$ is nonincreasing in λ ;
- RH is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ ;
- numerically $\lambda_{\min}(\lambda)$ is positive but tiny, and the corresponding ground state is close to a vector built from prolate spheroidal wave functions [3, 4].

Proving positivity window by window therefore requires control of a margin that shrinks very fast with λ . This paper quantifies that shrinking from above, unconditionally.

1.2. Results. Put $\mu = \lambda^2$.

Theorem A. *There are effectively computable constants C, p, λ_0 such that for all $\lambda \geq \lambda_0$*

$$\lambda_{\min}(\lambda) \leq C \lambda^p e^{-2\pi\lambda^2}.$$

More precisely, for $\mu \geq 5$ we have $\lambda_{\min}(\lambda) \leq 2S B(\lambda)^2/c_0^2$. Here:

- $B(\lambda)$ is the explicit elementary function (4.4), and $B(\lambda)e^{\pi\mu} = O(\mu^4)$;
- $S = \sum_{\rho} m_{\rho}(1 + |\gamma_{\rho}|^2)^{-1} < 0.046192$;
- $c_0^2 = 0.00894$ is a lower bound for $\|k\|^2$ (Lemma 4.2).

In closed form, for every $\mu \geq 5$, $\lambda_{\min}(\lambda) \leq 1.563 \times 10^9 \mu^8 e^{-2\pi\mu}$.

Theorem B. *There are constants C, p, λ_0 such that for all $\lambda \geq \lambda_0$*

$$\lambda_{\min}(\lambda) \leq C \lambda^p e^{-4\pi\lambda^2}.$$

One may take $p = 46$, and C can be made explicit; λ_0 is not effective.

Both statements are unconditional: no hypothesis on the location of the zeros of ζ is used. In terms of the half-width $a = \log \lambda$ of the window in logarithmic coordinates, $\mu = e^{2a}$, so the bounds decay doubly exponentially in a . The value $p = 46$ is not optimized; see Remark 2.

1.3. Relation to previous work.

- **The prolate heuristics.** Let $1 - \chi_2(\lambda)$ be the prolate eigenvalue defect defined in Section 6; by a theorem of Fuchs [9] it behaves like $\mu^{9/2} e^{-4\pi\mu}$. Connes, Consani and Moscovici [4, Fig. 4] and Connes [6, Fig. 1] observed that $\log \lambda_{\min}(\lambda)$ and $\log(1 - \chi_2(\lambda))$ behave strikingly alike as functions of μ ; the figure in [4] covers $2 \leq \mu \leq 17$. Theorem B proves the upper-bound half of this picture at the level of the exponent: $\lambda_{\min} \leq e^{-4\pi\mu + O(\log \mu)}$. Our own computations in Section 6 repeat the comparison for $\mu \leq 20$; they are not new in substance.
- **Zhu's conditional bound.** Zhu [18] proves, *assuming RH*, the bound $\lambda^*(L) \leq \exp(-Le^L)$ for windows $[-L, L]$ in logarithmic coordinates, so $L = a$ and $\exp(-Le^L) = \exp(-\frac{1}{2}\sqrt{\mu} \log \mu)$. Theorems A and B remove the hypothesis and replace $\sqrt{\mu} \log \mu$ by $2\pi\mu$ and $4\pi\mu$.
- **Zhu's certified numerics.** Zhu also gives certified *numerical* upper bounds for $\lambda^*(L)$ with $0.5 \leq L \leq 2$, evaluated unconditionally on the geometric side, together with an empirical law $-\ln \lambda^* \simeq 2\pi^2 N(T^*)/\ln N(T^*)$. What is new in Section 6 is the comparison of these numbers with $1 - \chi_2$. The ratio stays between 3 and 20 for $0.5 \leq L \leq 2$ (that is, $2.7 \leq \mu \leq 54.6$), while the numbers themselves range over 283 orders of magnitude. This extends the observation of [4, 6] to $\mu \approx 55$ and is consistent with $-\ln \lambda_{\min} = 4\pi\mu + O(\log \mu)$. Zhu's fitted law would instead behave like $2\pi^2\mu$ as $\mu \rightarrow \infty$. This is an observation about upper bounds only; the true asymptotics can be decided only with lower bounds, which are of the same strength as RH.
- **Positivity on a fixed window.** In earlier work [12] the author proved $W(f) > 0$ on the window $[-\frac{1}{2} \log 5, \frac{1}{2} \log 5]$ with a Lean formalization. The present paper goes in the opposite direction: it bounds from above, unconditionally, the margin that any such window-by-window positivity proof can have.
- **The CCM program.** In [4] two steps remain open in the program:
 - (1) the ground state is simple and even;
 - (2) the ground state is close to the vector k_{λ} built from prolate functions.
 Our results concern neither step directly. Zhu [18, Thm. 6.2] certifies step 1 for the single window $L = 0.8$.

1.4. Idea of the proof. Let $\mathcal{E}(\varphi)(u) = u^{1/2} \sum_{n \geq 1} \varphi(nu)$. If φ is even and $\varphi(0) = \int \varphi = 0$, then the Mellin transform of $\mathcal{E}(\varphi)$ equals $\zeta(\frac{1}{2} + iz)$ times a holomorphic function. Hence $\mathcal{E}(\varphi)$ lies in the radical of the Weil form: its Fourier transform vanishes at every zero ordinate γ_{ρ} , whether or not ρ is on the critical line. This fact is classical [5, 3].

Now suppose φ is supported in $[-\lambda, \lambda]$. Then $\mathcal{E}(\varphi)$ vanishes for $u \geq \lambda$. Its restriction k to the window differs from $\mathcal{E}(\varphi)$ only by the piece r on $u < \lambda^{-1}$. By Poisson summation this piece is

controlled by $\hat{\varphi}$ on $|\xi| \geq \lambda$. Since $\hat{k} = -\hat{r}$ at the zeros, the explicit formula gives

$$|W(k)| \leq 2S \left(\text{size of } \hat{\varphi} \text{ beyond } \lambda \right)^2.$$

Theorem A takes φ to be Riemann's function h , whose Fourier transform is itself and decays like $e^{-\pi\xi^2}$, cut off smoothly. Theorem B takes a Kaiser–Bessel window, whose Fourier transform beyond the band edge is smaller by a further factor $e^{-2\pi\lambda^2}$. This is the familiar concentration property of time- and band-limited functions [14].

1.5. Formalization. Theorem A is formalized in Lean 4 [11, 13] as a statement about the geometric side of the explicit formula (Section 7). It builds on the formalization of Weil's explicit formula in the Lean library Zeta23 [17], which is released under the Apache License 2.0.

2. SETTING AND CONVENTIONS

Let $\lambda > 1$ and $a = \log \lambda$. In the variable $y = \log u$, the space $L^2([\lambda^{-1}, \lambda], d^*u)$ becomes $L^2([-a, a], dy)$. We use two Fourier transforms:

$$\hat{f}(z) = \int_{\mathbb{R}} f(y) e^{izy} dy \quad (\text{logarithmic variable}), \quad \hat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(x) e^{2\pi i x \xi} dx \quad (\text{additive variable}).$$

For a nontrivial zero ρ of ζ with multiplicity m_ρ , put $\gamma_\rho = (\rho - \frac{1}{2})/i$. Then $|\text{Im } \gamma_\rho| < \frac{1}{2}$, and the multiset of the γ_ρ is invariant under $\gamma \mapsto -\gamma$ and $\gamma \mapsto \bar{\gamma}$.

Weil's quadratic form $W(f) = QW(f, f)$ is given on the geometric side as in [4, (3.19)]. Let $\lambda_{\min}(\lambda)$ be the bottom of the spectrum of its restriction to $L^2([-a, a])$ [4, Thm. 3.6, Cor. 3.7]. Piecewise smooth functions supported in $[-a, a]$ belong to the form domain [4, after (3.27)]. Hence

$$(2.1) \quad \lambda_{\min}(\lambda) \leq W(f)/\|f\|^2 \quad \text{for every such } f \neq 0.$$

Finally put $S = \sum_{\rho} m_\rho (1 + |\gamma_\rho|^2)^{-1}$.

3. A MASTER PROPOSITION

Definition 3.1 (Hypothesis (H)). A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ satisfies (H) if:

- φ is even, C^1 , and piecewise C^2 with finitely many break points;
- $\text{supp } \varphi \subset [-\lambda, \lambda]$ and $\varphi(\pm\lambda) = 0$;
- $\varphi(v) = O(v^2)$ as $v \rightarrow 0$, and $\int \varphi = 0$;
- there are $\alpha, \alpha' > 1$ and C_φ with $|\hat{\varphi}(\xi)| \leq C_\varphi (1 + |\xi|)^{-\alpha}$ and $|\xi \hat{\varphi}'(\xi)| \leq C_\varphi (1 + |\xi|)^{-\alpha'}$.

Given such φ , put

$$e(y) = e^{y/2} \sum_{n \geq 1} \varphi(ne^y), \quad k = e \mathbf{1}_{[-a, a]}, \quad r = e \mathbf{1}_{(-\infty, -a]}.$$

Thus $e(\log u) = \mathcal{E}(\varphi)(u)$. Near any $y_0 < a$ only finitely many n contribute, so e is C^1 on $(-\infty, a)$. For $y \geq a$ every term vanishes, because $ne^y \geq \lambda$. Since $\varphi(\lambda) = \varphi'(\lambda) = 0$, e is C^1 on $(-\infty, \infty)$ with $e = 0$ on $[a, \infty)$, and $e = k + r$. In particular k is piecewise C^1 , with a single jump, at $-a$.

Lemma 3.2 (Explicit formula). *Let f be real, compactly supported and piecewise C^1 with finitely many jumps. Then*

$$W(f) = \sum_{\rho} m_\rho \hat{f}(\gamma_\rho) \hat{f}(-\gamma_\rho),$$

and the sum converges absolutely.

Proof. For $f \in C_c^\infty$ this is Weil's explicit formula for $g = f * f^*$, where $f^*(y) = f(-y)$ and $\hat{g}(z) = \hat{f}(z)\hat{f}(-z)$ [16, 2]. It is unconditional.

In general the distributional derivative f' is a finite measure, so integration by parts gives $|\hat{f}(z)| \leq C_f/(1+|z|)$ on the strip $|\operatorname{Im} z| \leq \frac{1}{2}$. Let $f_n = f * \varrho_{1/n}$ with an even mollifier $\varrho_{1/n} \geq 0$ supported in $[-\frac{1}{n}, \frac{1}{n}]$. Then $|\hat{\varrho}_{1/n}| \leq e^{1/(2n)}$ on the strip, so the bound $|\hat{f}_n(z)| \leq e^{1/2}C_f/(1+|z|)$ holds with one constant, and $\hat{f}_n \rightarrow \hat{f}$ pointwise. On the zero side we use dominated convergence, since $\sum_\rho m_\rho(1+|\gamma_\rho|^2)^{-1} < \infty$. On the geometric side [4, (3.19)]:

- the archimedean term $\int |\hat{f}_n(t)|^2 \partial_t \theta(t) dt / \pi$ converges by dominated convergence, since $\partial_t \theta(t) = O(\log(2+|t|))$ [4, (3.24)] and $|\hat{f}_n(t)| \leq |\hat{f}(t)| = O(1/(1+|t|))$ on $\mathbb{R}$;
- the pole terms $\hat{f}_n(\pm \frac{i}{2})\hat{f}_n(\mp \frac{i}{2})$ converge;
- the prime sum is a finite sum of values of $f_n * f_n^*$, which converge uniformly.

Passing to the limit on both sides gives the claim. $\square$

Lemma 3.3 (Poisson summation). *Assume (H). For every $y \in \mathbb{R}$,*

$$(3.1) \quad e(y) = e^{-y/2} \sum_{m \geq 1} \hat{\varphi}(me^{-y}),$$

and for $y \leq 0$,

$$|e(y)| \leq C_\varphi \zeta(\alpha) e^{(\alpha - \frac{1}{2})y}, \quad |e'(y)| \leq \frac{1}{2}|e(y)| + C_\varphi \zeta(\alpha') e^{(\alpha' - \frac{1}{2})y}.$$

Proof. Fix $x > 0$. The function $u \mapsto \sum_{n \in \mathbb{Z}} \varphi((n+u)x)$ is continuous and 1-periodic, and its Fourier coefficients are $x^{-1}\hat{\varphi}(m/x)$ up to sign of the index. They are absolutely summable because $\alpha > 1$. Hence the Fourier series converges to the function everywhere, and at $u = 0$

$$\sum_{n \in \mathbb{Z}} \varphi(nx) = x^{-1} \sum_{m \in \mathbb{Z}} \hat{\varphi}(m/x).$$

Since φ is even and $\varphi(0) = \hat{\varphi}(0) = \int \varphi = 0$, both sides are twice the sums over $n, m \geq 1$. Put $x = e^y$ and multiply by $e^{y/2}$; this gives (3.1).

For $y \leq 0$ we have $|\hat{\varphi}(me^{-y})| \leq C_\varphi m^{-\alpha} e^{\alpha y}$, which gives the bound on e . The series differentiated termwise,

$$e'(y) = -\frac{1}{2}e(y) - e^{-y/2} \sum_{m \geq 1} (me^{-y})\hat{\varphi}'(me^{-y}),$$

converges uniformly on $(-\infty, 0]$ by the bound on $\xi\hat{\varphi}'$, which justifies the differentiation and gives the bound on e' . $\square$

Lemma 3.4 (Radical). *Assume (H). Then $\hat{e}$ is holomorphic on $\operatorname{Im} z < \alpha - \frac{1}{2}$, and $\hat{e}(\gamma_\rho) = \hat{e}(-\gamma_\rho) = 0$ for every nontrivial zero ρ .*

Proof. By Lemma 3.3 and $e = 0$ on $[a, \infty)$, the integral $\hat{e}(z) = \int_{-\infty}^a e(y)e^{izy}dy$ converges locally uniformly on $\operatorname{Im} z < \alpha - \frac{1}{2}$, so $\hat{e}$ is holomorphic there. This half-plane contains the strip $|\operatorname{Im} z| \leq \frac{1}{2}$.

Put $\Phi(z) = \int_0^\infty \varphi(v)v^{-\frac{1}{2}+iz}dv$. It is holomorphic on $\operatorname{Im} z < \frac{5}{2}$, because $\varphi = O(v^2)$ at 0 and φ has compact support. For $\operatorname{Im} z < -\frac{1}{2}$, Fubini's theorem is justified by $\sum_n n^{-(\frac{1}{2}-\operatorname{Im} z)} < \infty$, and the substitution $v = ne^y$ gives

$$\hat{e}(z) = \sum_{n \geq 1} \int_{\mathbb{R}} \varphi(ne^y)e^{(\frac{1}{2}+iz)y}dy = \sum_{n \geq 1} n^{-\frac{1}{2}-iz}\Phi(z) = \zeta(\frac{1}{2}+iz)\Phi(z).$$

Both sides are holomorphic on the connected open set $D = \{\operatorname{Im} z < \alpha - \frac{1}{2}\} \setminus \{-\frac{i}{2}\}$, and they agree on an open subset of D . By the identity theorem they agree on D . Since $\rho \neq 1$, we have $\gamma_\rho \in D$ and $\hat{e}(\gamma_\rho) = \zeta(\rho)\Phi(\gamma_\rho) = 0$. Likewise $-\gamma_\rho = \gamma_{1-\rho}$ and $\zeta(1-\rho) = 0$. Only the equation $\zeta(\rho) = 0$ is used, not the location of ρ . $\square$

Proposition 3.5 (Zero-side estimate). *Put*

$$A_0 = \int_{-\infty}^{-a} |r(y)| e^{|y|/2} dy, \quad A_1 = |r(-a)| e^{a/2} + \int_{-\infty}^{-a} |r'(y)| e^{|y|/2} dy.$$

Then

$$|W(k)| \leq 2(A_0 + A_1)^2 S, \quad S \leq 2\left(1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi\right) < 0.046192.$$

Proof. By Lemma 3.2 applied to k , and by Lemma 3.4 with $e = k + r$, we have $\hat{k}(\pm\gamma_\rho) = -\hat{r}(\pm\gamma_\rho)$ and hence

$$W(k) = \sum_{\rho} m_{\rho} \hat{r}(\gamma_{\rho}) \hat{r}(-\gamma_{\rho}).$$

On the strip $|\operatorname{Im} z| \leq \frac{1}{2}$ and for $y < 0$ we have $|e^{izy}| \leq e^{|y|/2}$, so $|\hat{r}(z)| \leq A_0$. Integrating by parts once,

$$iz \hat{r}(z) = r(-a) e^{-iza} - \int_{-\infty}^{-a} r'(y) e^{izy} dy,$$

where the boundary term at $-\infty$ vanishes because $|r(y)| e^{|y|/2} = O(e^{(\alpha-1)y})$ and $\alpha > 1$. Hence $|z \hat{r}(z)| \leq A_1$. Now $\min(A_0, A_1/|z|)^2 \leq 2(A_0 + A_1)^2/(1 + |z|^2)$: for $|z| \leq 1$ use the first entry, and for $|z| > 1$ the second. Since $|\gamma_{\rho}| = |\gamma_{\rho}|$, this gives $|W(k)| \leq 2(A_0 + A_1)^2 S$.

For the bound on S , write $\rho = \beta + it$; then $|\gamma_{\rho}|^2 \geq t^2$. The map $\rho \mapsto 1 - \bar{\rho}$ is an involution on the zeros that preserves multiplicities. If $\beta \neq \frac{1}{2}$, then

$$\operatorname{Re} \frac{1}{\rho} + \operatorname{Re} \frac{1}{1 - \bar{\rho}} = \frac{\beta}{\beta^2 + t^2} + \frac{1 - \beta}{(1 - \beta)^2 + t^2} \geq \frac{1}{1 + t^2},$$

since $\beta^2, (1 - \beta)^2 \leq 1$; the pair contributes at most $2/(1 + t^2)$ to S . If $\beta = \frac{1}{2}$, then $\operatorname{Re}(1/\rho) \geq \frac{1}{2}(1 + t^2)^{-1}$. Hence $S \leq 2 \sum_{\rho} m_{\rho} \operatorname{Re}(1/\rho)$, and the classical unconditional identity $\sum_{\rho} m_{\rho} \operatorname{Re}(1/\rho) = 1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi = 0.0230957 \dots$ [7, Ch. 12] gives the bound. $\square$

Lemma 3.6 (Poisson form of A_0, A_1). *Let $T_0(s) = \int_s^{\infty} |\hat{\varphi}(t)| dt$ and $T_1(s) = \int_s^{\infty} t |\hat{\varphi}'(t)| dt$. Then*

$$A_0 + A_1 \leq B_{\varphi}(\lambda) := \frac{3}{2} \sum_{m \geq 1} \frac{T_0(m\lambda)}{m} + \sum_{m \geq 1} \frac{T_1(m\lambda)}{m} + \lambda \sum_{m \geq 1} |\hat{\varphi}(m\lambda)|.$$

Proof. Put $x = e^{-y} \in (\lambda, \infty)$ for $y < -a$. Then $e^{|y|/2} = x^{1/2}$, $|dy| = dx/x$, and $r = x^{1/2} \sum_m \hat{\varphi}(mx)$ by (3.1). Hence

$$A_0 \leq \sum_{m \geq 1} \int_{\lambda}^{\infty} |\hat{\varphi}(mx)| dx = \sum_{m \geq 1} \frac{T_0(m\lambda)}{m}.$$

Next, $r'(y) = -x \frac{d}{dx} (x^{1/2} \sum_m \hat{\varphi}(mx))$, so

$$|r'(y)| e^{|y|/2} |dy| \leq \left(\frac{1}{2} \sum_m |\hat{\varphi}(mx)| + \sum_m |mx \hat{\varphi}'(mx)| \right) dx,$$

and $\int_{\lambda}^{\infty} |mx \hat{\varphi}'(mx)| dx = T_1(m\lambda)/m$. Finally $|r(-a)| e^{a/2} = \lambda |\sum_m \hat{\varphi}(m\lambda)|$. $\square$

Proposition 3.7 (Master proposition). *Assume (H). If $\|k\|^2 \geq c_0^2 > 0$, then $\lambda_{\min}(\lambda) \leq 2S B_{\varphi}(\lambda)^2 / c_0^2$.*

Proof. By (2.1), $\lambda_{\min} \leq W(k)/\|k\|^2 \leq |W(k)|/\|k\|^2$. Now apply Proposition 3.5 and Lemma 3.6. The absolute value is needed because, without RH, $W(k)$ could be negative. $\square$

4. PROOF OF THEOREM A

4.1. Construction. Let $h(x) = \frac{\pi}{2}x^2(2\pi x^2 - 3)e^{-\pi x^2}$. It is a combination of the Hermite functions of degrees 0 and 4, so $\hat{h} = h$. Moreover $h(0) = \int h = 0$, and $|h|$ is decreasing on $[0.977, \infty)$.

Put $\delta = \lambda^{-2}$ and $b = \lambda - 2\delta$. Let $S_5(t) = 10t^3 - 15t^4 + 6t^5$ be the C^2 smoothstep, with

$$M_1 = \sup |S'_5| = \frac{15}{8}, \quad M_2 = \sup |S''_5| = \frac{10}{\sqrt{3}}, \quad M_3 = \sup |S'''_5| = 60.$$

Let χ be 1 on $|x| \leq b$, equal to $1 - S_5((|x| - b)/\delta)$ on $b \leq |x| \leq b + \delta$, and 0 beyond. Let $\psi = \frac{315}{32}x^2(1 - x^2)^3 \mathbf{1}_{|x| \leq 1}$; it is C^2 , with $\psi(0) = 0$ and $\int \psi = 1$. Set

$$\varphi_A = h\chi + \varepsilon\psi, \quad \varepsilon = \int h(1 - \chi), \quad g = h(1 - \chi).$$

Lemma 4.1. φ_A satisfies (H) with $\alpha = \alpha' = 2$. More precisely, for $t > 0$,

$$(4.1) \quad |\hat{\varphi}_A(t)| \leq |h(t)| + \frac{D_0}{(2\pi t)^2}, \quad |t\hat{\varphi}'_A(t)| \leq |th'(t)| + \frac{D_1}{(2\pi t)^2},$$

where $D_0 = \|g''\|_1 + |\varepsilon| \|\psi''\|_1$ and $D_1 = \|(xg)'''\|_1 + |\varepsilon| \|(x\psi)'''\|_1$.

Proof. Evenness and $\varphi_A \in C^2$ are clear, and the support lies in $[-(\lambda - \delta), \lambda - \delta]$. Near 0, $\varphi_A = h + \varepsilon\psi = O(v^2)$. Also $\int \varphi_A = \int h\chi + \varepsilon = \int h = 0$.

Since $h\chi = h - g$, we have $\hat{\varphi}_A = h - \hat{g} + \varepsilon\hat{\psi}$. The function g is C^2 , g'' is absolutely continuous and $g''' \in L^1$. Two integrations by parts give $|\hat{g}(t)| \leq \|g''\|_1/(2\pi t)^2$. Moreover $t\hat{g}'(t) = -(\widehat{xg})'(t)$, and two further integrations by parts give $|t\hat{g}'(t)| \leq \|(xg)'''\|_1/(2\pi t)^2$. The same holds for ψ . $\square$

4.2. Explicit tails. Write $h^{(i)} = Q_i e^{-\pi x^2}$ with polynomials $Q_i = \sum_k q_{ik} x^k$, and put $U_i(b) = \sum_k |q_{ik}| b^k e^{-\pi b^2}$. Each $x^k e^{-\pi x^2}$ decreases for $x \geq \sqrt{k/2\pi}$, so $U_i(b) \geq \sup_{x \geq b} |h^{(i)}(x)|$ for $b \geq 1.13$ and $i \leq 3$. For the Gaussian tails we use

$$(4.2) \quad \int_s^\infty t^k e^{-\pi t^2} dt \leq \frac{s^{k-1} e^{-\pi s^2}}{2\pi(1 - \frac{k-1}{2\pi s^2})} \quad (k \geq 1), \quad \int_s^\infty e^{-\pi t^2} dt \leq \frac{e^{-\pi s^2}}{2\pi s},$$

the first by one integration by parts. With $\chi' = O(M_1/\delta)$, $\chi'' = O(M_2/\delta^2)$, $\chi''' = O(M_3/\delta^3)$ on two intervals of length δ , the Leibniz rule gives

$$(4.3) \quad \begin{aligned} \|g''\|_1 &\leq 2 \left(\int_b^\infty |h''| + 2M_1 U_1(b) + M_2 U_0(b)/\delta \right), \\ \|(xg)'''\|_1 &\leq 2 \left(\int_b^\infty x|h'''\| + \lambda(3M_1 U_2(b) + 3M_2 U_1(b)/\delta + M_3 U_0(b)/\delta^2) \right) + 3\|g''\|_1, \\ |\varepsilon| &\leq 2 \int_b^\infty |h|, \quad \|\psi''\|_1 = 27.194\dots, \quad \|(x\psi)'''\|_1 = 158.23\dots \end{aligned}$$

Here we used $(xg)''' = xg''' + 3g''$ and $b + \delta \leq \lambda$. The integrals $\int_b^\infty |h^{(i)}|$ and $\int_b^\infty x|h'''\|$ are bounded termwise by (4.2). Inserting (4.1) into Lemma 3.6 and using $\sum_m m^{-2} = \zeta(2)$ gives $B_{\varphi_A}(\lambda) \leq B(\lambda)$, where

$$(4.4) \quad B(\lambda) := \sum_{m \geq 1} \left[\frac{3}{2m} \int_{m\lambda}^\infty |h| + \frac{1}{m} \int_{m\lambda}^\infty t|h'(t)| dt + \lambda|h(m\lambda)| \right] + \frac{\zeta(2)(\frac{5}{2}D_0 + D_1)}{4\pi^2\lambda},$$

with D_0, D_1 bounded by (4.3). Each bracket is at most $P(m\lambda)e^{-\pi m^2 \lambda^2}$ with P a polynomial of degree at most 6, so the terms with $m \geq 60$ contribute less than 10^{-10^4} for $\mu \geq 5$; in the computations we sum $m \leq 59$ and add this bound.

As a check of the direction of the inequalities, B exceeds the value of $A_0 + A_1$ computed from the definitions for the same φ_A by factors 140, 110, 86, 80 at $\mu = 5, 7, 11, 13$.

4.3. Lower bound for $\|k\|$.

Lemma 4.2. *For $\mu \geq 5$, $\|k\|^2 \geq c_0^2 := 0.00894$.*

Proof. Since $a \geq \frac{1}{2} \log 5 > 0.8$, we have $\|k\|^2 \geq \int_{-\eta}^{\eta} |e|^2$ with $\eta = 0.1$. Let $e_h(y) = e^{y/2} \sum_n h(ne^y)$. Ball arithmetic (Arb [1], 400 subintervals, with the tail of the sum over n bounded explicitly) gives $\int_{-\eta}^{\eta} e_h^2 \geq 0.0093256$.

On $[-\eta, \eta]$ we have $e - e_h = e^{y/2} \sum_n (\varepsilon\psi - g)(ne^y)$. Only $n = 1$ meets $\text{supp } \psi$, since $2e^{-\eta} > 1$. The function g vanishes on $[0, b]$ and $|g| \leq |h|$. The points $ne^y \geq b$ have spacing $e^y \geq e^{-\eta}$, and $|h|$ decreases on $[b, \infty)$, so the first such point contributes at most $U_0(b)$ and the others at most $e^\eta \int_b^\infty |h|$. Hence $|e - e_h| \leq E$ on $[-\eta, \eta]$, where

$$E = e^{\eta/2} \left(|\varepsilon| \sup \psi + U_0(b) + e^\eta \int_b^\infty |h| \right).$$

By Minkowski's inequality, $\|k\| \geq \sqrt{0.0093256} - \sqrt{2\eta} E$. At $\mu = 5$ we have $E \leq 4.5 \times 10^{-3}$, which gives $\|k\|^2 \geq 0.00894$. Since b increases with μ , E decreases, and the bound holds for all $\mu \geq 5$. $\square$

4.4. Conclusion. Proposition 3.7, Lemma 4.1, the bound $B_{\varphi_A} \leq B$ and Lemma 4.2 give $\lambda_{\min}(\lambda) \leq 2SB(\lambda)^2/c_0^2$ for $\mu \geq 5$.

Every term of $B(\lambda)e^{\pi\lambda^2}$ is at most a polynomial in λ of degree at most 8 times $e^{\pi(\lambda^2-b^2)}$, and $\lambda^2 - b^2 \leq 4/\lambda$. The top degree comes from the term $\lambda M_3 U_0(b)/\delta^2$ in D_1 , divided by λ in (4.4). Hence $B(\lambda) \leq K\mu^4 e^{-\pi\mu}$ with an effective K , and $\lambda_{\min} \leq C\mu^8 e^{-2\pi\mu}$.

For the closed form, put $F(\lambda) = B(\lambda)e^{\pi\mu}/\mu^4$. We show $F \leq 12298$ for all $\mu \geq 5$.

- On $5 \leq \mu \leq 10^4$ we evaluate F in ball arithmetic on 48,882 intervals in λ . The Gaussian factors are combined with $e^{\pi\lambda^2}$ into $e^{-\pi(s^2-\lambda^2)}$, so that the enclosures do not blow up. The coefficients q_{ik} and the norms of ψ are computed exactly. The maximum is attained near $\mu = 5$, where the point value is 12163.95.
- For $\lambda \geq 100$, replace b by λ in the polynomials, bound $e^{\pi(\lambda^2-b^2)} \leq e^{4\pi/100}$ and the factors $(1 - \frac{k-1}{2\pi s^2})^{-1}$ by their values at $s = 99.9$. This turns the $m = 1$ part of $B(\lambda)e^{\pi\lambda^2}$ into a Laurent polynomial in λ with nonnegative coefficients and top degree 8, so its quotient by λ^8 is nonincreasing, and at $\lambda = 100$ it is at most 57.0. The terms with $m \geq 2$ contribute less than 10^{-10^4} .

With $S \leq 0.046192$ and $c_0^2 = 0.00894$, this gives $2S \cdot 12298^2/c_0^2 \leq 1.563 \times 10^9$. This completes the proof of Theorem A.

5. PROOF OF THEOREM B

5.1. Construction. Let $\beta = 2\pi\lambda^2$. For $\alpha > -1$ put

$$\mathfrak{F}_\alpha(z) = \sum_{j \geq 0} \frac{(z/4)^j}{2^\alpha j! \Gamma(j + \alpha + 1)},$$

an entire function of z . Then $\mathfrak{F}_\alpha(w^2) = I_\alpha(w)/w^\alpha$ and $\mathfrak{F}_\alpha(-w^2) = J_\alpha(w)/w^\alpha$ for $w > 0$, and termwise differentiation gives

$$(5.1) \quad \frac{d}{d\omega} \mathfrak{F}_\alpha(\beta^2 - \omega^2) = -\omega \mathfrak{F}_{\alpha+1}(\beta^2 - \omega^2).$$

Let

$$\Phi_\beta(t) = (1 - t^2) I_2(\beta \sqrt{1 - t^2}) \mathbf{1}_{|t| \leq 1}, \quad \varphi_B(x) = \frac{x^2(x^2 - \sigma) \Phi_\beta(x/\lambda)}{I_2(\beta)},$$

where $\sigma = \int x^4 \Phi_\beta(x/\lambda) dx / \int x^2 \Phi_\beta(x/\lambda) dx$, so that $\int \varphi_B = 0$. Note $0 < \sigma \leq \lambda^2$. Since $I_2(z) = z^2 E(z^2)$ with E entire, $\Phi_\beta(t) = \beta^2(1 - t^2)^2 E(\beta^2(1 - t^2))$ on $[-1, 1]$.

Lemma 5.1 (Sonine–Gegenbauer). *For $\nu \geq 0$ and $\omega \in \mathbb{R}$,*

$$\int_{-1}^1 (1-t^2)^{\nu/2} I_\nu(\beta\sqrt{1-t^2}) e^{i\omega t} dt = \sqrt{2\pi} \beta^\nu \mathfrak{F}_{\nu+\frac{1}{2}}(\beta^2 - \omega^2).$$

Proof. Expand I_ν in its power series and integrate termwise, using $\int_{-1}^1 (1-t^2)^m e^{i\omega t} dt = \sqrt{\pi} \Gamma(m+1)(2/\omega)^{m+\frac{1}{2}} J_{m+\frac{1}{2}}(\omega)$ and the series of $J_{m+\frac{1}{2}}$; the resulting double series rearranges into the series of $\mathfrak{F}_{\nu+\frac{1}{2}}$. The formula is classical [15, Ch. XII]. We also checked it numerically at 18 points, with relative error at most 3×10^{-17} . $\square$

Lemma 5.2 (Bessel bounds). (i) $|J_\alpha(w)| \leq 1$ for $\alpha \geq 0$, and $|J_\alpha(w)| \leq (w/2)^\alpha / \Gamma(\alpha+1)$ for $\alpha \geq -\frac{1}{2}$, $w \geq 0$ [8, 10.14.1, 10.14.4]. Consequently $|\mathfrak{F}_\alpha(-w^2)| \leq \min(2^{-\alpha} \Gamma(\alpha+1)^{-1}, w^{-\alpha})$.
(ii) For $\nu \geq \frac{1}{2}$ and $z > 0$, $I_\nu(z) \leq e^z / \sqrt{2\pi z}$.
(iii) For $\nu \geq \frac{1}{2}$ and $z \geq 1$, $I_\nu(z) \geq c_\nu e^z / \sqrt{z}$, with $c_\nu = e^{-1} 2^{-\nu} / (\sqrt{\pi} \Gamma(\nu + \frac{3}{2}))$. In particular $c_2 = 0.015613\dots$

Proof. For (ii) and (iii) use the Poisson integral

$$I_\nu(z) = \frac{(z/2)^\nu}{\sqrt{\pi} \Gamma(\nu + \frac{1}{2})} \int_{-1}^1 (1-t^2)^{\nu-\frac{1}{2}} e^{zt} dt.$$

For (ii), $(1-t^2)^{\nu-\frac{1}{2}} \leq 2^{\nu-\frac{1}{2}} (1-t)^{\nu-\frac{1}{2}}$ and $\int_{-1}^1 (1-t)^{\nu-\frac{1}{2}} e^{zt} dt \leq e^z \Gamma(\nu + \frac{1}{2}) z^{-\nu-\frac{1}{2}}$. For (iii), restrict to $t \in [1-1/z, 1]$; there $(1-t^2)^{\nu-\frac{1}{2}} \geq (1-t)^{\nu-\frac{1}{2}}$ and $e^{zt} \geq e^{z-1}$, and $\int_0^{1/z} u^{\nu-\frac{1}{2}} du = z^{-\nu-\frac{1}{2}} / (\nu + \frac{1}{2})$. $\square$

Numerically, for $\nu = 2$, $\max_{0 < z \leq 2000} I_2(z) \sqrt{2\pi z} e^{-z} = 0.99906$ and $\min_{z \geq 1} I_2(z) \sqrt{z} e^{-z} = 0.04994$.

5.2. The Fourier transform of φ_B . Put $t = x/\lambda$ and $\omega = 2\pi\lambda\xi$, so that $|\omega| \geq \beta$ if and only if $|\xi| \geq \lambda$. Multiplication by t^2 corresponds to $-\partial_\omega^2$. By Lemma 5.1 with $\nu = 2$,

$$(5.2) \quad \hat{\varphi}_B(\xi) = \frac{\lambda}{I_2(\beta)} (\lambda^4 \partial_\omega^4 + \sigma \lambda^2 \partial_\omega^2) G(\omega), \quad G(\omega) = \sqrt{2\pi} \beta^2 \mathfrak{F}_{5/2}(\beta^2 - \omega^2).$$

By (5.1), each ω -derivative either lowers the power of ω by one, or raises it by one and the index of $\mathfrak{F}$ by one. Hence $\partial_\omega^j G$ is a linear combination, with integer coefficients depending only on j , of the terms $\omega^{2l-j} \mathfrak{F}_{\frac{5}{2}+l}(\beta^2 - \omega^2)$ with $j/2 \leq l \leq j$. Since $\xi \partial_\xi = \omega \partial_\omega$ maps $\omega^i \mathfrak{F}_{\frac{5}{2}+l}$ to $i\omega^i \mathfrak{F}_{\frac{5}{2}+l} - \omega^{i+2} \mathfrak{F}_{\frac{7}{2}+l}$, the function $\xi \hat{\varphi}'_B(\xi)$ is a combination of terms $\omega^i \mathfrak{F}_{\frac{5}{2}+l}$ with $i \leq l+1$ and $l \leq 5$, and $\hat{\varphi}_B$ one of terms with $i \leq l \leq 4$. Explicitly, writing $\mathfrak{F}_\alpha$ for $\mathfrak{F}_\alpha(\beta^2 - \omega^2)$,

$$\frac{G'''}{\sqrt{2\pi}\beta^2} = -\mathfrak{F}_{\frac{7}{2}} + \omega^2 \mathfrak{F}_{\frac{9}{2}}, \quad \frac{G''''}{\sqrt{2\pi}\beta^2} = 3\mathfrak{F}_{\frac{9}{2}} - 6\omega^2 \mathfrak{F}_{\frac{11}{2}} + \omega^4 \mathfrak{F}_{\frac{13}{2}},$$

and $\omega \partial_\omega$ of the last term is $4\omega^4 \mathfrak{F}_{\frac{13}{2}} - \omega^6 \mathfrak{F}_{\frac{15}{2}}$, which is the term with the largest power $i = 6$.

Lemma 5.3. φ_B satisfies (H) with $(\alpha, \alpha') = (\frac{5}{2}, \frac{3}{2})$. Moreover, for $|\xi| \geq \lambda$ and $\lambda \geq 1$,

$$(5.3) \quad |\hat{\varphi}_B(\xi)| \leq C \lambda^{18} e^{-2\pi\lambda^2} (\lambda/|\xi|)^{5/2}, \quad |\xi \hat{\varphi}'_B(\xi)| \leq C \lambda^{22} e^{-2\pi\lambda^2} (\lambda/|\xi|)^{3/2},$$

with an explicit absolute constant C .

Proof. Evenness, $\varphi_B = O(v^2)$ and $\int \varphi_B = 0$ are clear. By the formula for Φ_β above, φ_B is real-analytic on $(-\lambda, \lambda)$ and vanishes to second order at $\pm\lambda$; so it is C^1 , piecewise C^∞ , and $\varphi_B(\pm\lambda) = 0$.

For $|\xi| \geq \lambda$ every $\mathfrak{F}$ in (5.2) is evaluated at $-w^2 \leq 0$, where $w = \sqrt{\omega^2 - \beta^2}$. By Lemma 5.2(i):

- for $\beta \leq |\omega| \leq 2\beta$, $|\omega^i \mathfrak{F}_{\frac{5}{2}+l}(-w^2)| \leq (2\beta)^i$, with $i \leq 4$ for the terms of $\hat{\varphi}_B$ and $i \leq 6$ for those of $\xi \hat{\varphi}'_B$;

- for $|\omega| \geq 2\beta$, $w \geq \frac{\sqrt{3}}{2}|\omega|$ and $|\omega^i \mathfrak{F}_{\frac{5}{2}+l}(-w^2)| \leq (2/\sqrt{3})^{\frac{5}{2}+l} |\omega|^{i-l-\frac{5}{2}}$, which is at most a constant times $|\omega|^{-5/2}$ for the terms of $\hat{\varphi}_B$ ($i \leq l$) and $|\omega|^{-3/2}$ for the terms of $\xi \hat{\varphi}'_B$ ($i \leq l+1$).

In the first range $(\lambda/|\xi|)^{5/2} \geq 2^{-5/2}$; in the second $|\omega|^{-\kappa} = (2\pi\lambda^2)^{-\kappa} (\lambda/|\xi|)^{\kappa}$. Since $\sigma \leq \lambda^2$, the prefactor in (5.2) is at most $2\lambda^5 \sqrt{2\pi} \beta^2 / I_2(\beta)$, and $1/I_2(\beta) \leq c_2^{-1} \sqrt{\beta} e^{-\beta}$ by Lemma 5.2(iii). The prefactor is therefore $O(\lambda^{10} e^{-2\pi\lambda^2})$, and $(2\beta)^4 = O(\lambda^8)$, $(2\beta)^6 = O(\lambda^{12})$; this gives (5.3). For $|\xi| \leq \lambda$, $\hat{\varphi}_B$ and $\xi \hat{\varphi}'_B$ are bounded because φ_B and $x\varphi_B$ are integrable, so (H) holds. $\square$

Numerically, the true decay rates of $\hat{\varphi}_B$ and $\xi \hat{\varphi}'_B$ are $|\xi|^{-3}$ and $|\xi|^{-2}$ (checked up to $\xi = 1280\lambda$), and $\sup_{\lambda \leq \xi \leq 5\lambda} (|\hat{\varphi}_B| + |\xi \hat{\varphi}'_B|) e^{2\pi\mu}$ grows like $\mu^{11.06}$ for $\mu = 5-17$. More precisely, for $\mu = 5, 7, \dots, 17$ the quotients of $\sup_{\lambda \leq \xi \leq 5\lambda} |\hat{\varphi}_B(\xi)| e^{2\pi\mu} (\xi/\lambda)^{5/2}$ by λ^{18} and of $\sup_{\lambda \leq \xi \leq 5\lambda} |\xi \hat{\varphi}'_B(\xi)| e^{2\pi\mu} (\xi/\lambda)^{3/2}$ by λ^{22} stay in $[5.67, 5.80]$ and $[13.8, 15.2]$; the maxima are attained at the band edge $\xi = \lambda$. The powers in (5.3) cannot be lowered, as the following lower bound shows.

Remark 5.4 (Lower bound at the band edge). At $\xi = \lambda$ we have $\omega = \beta$, so every $\mathfrak{F}_a$ is evaluated at 0, where $\mathfrak{F}_a(0) = 1/(2^a \Gamma(a+1)) > 0$. The formulas for G''' and G'''' above give

$$\xi \hat{\varphi}'_B(\xi)|_{\xi=\lambda} = \frac{\lambda \sqrt{2\pi} \beta^2}{I_2(\beta)} \left[\lambda^4 \left(-\beta^6 \mathfrak{F}_{\frac{15}{2}} + 10\beta^4 \mathfrak{F}_{\frac{13}{2}} - 15\beta^2 \mathfrak{F}_{\frac{11}{2}} \right) + \sigma \lambda^2 \left(-\beta^4 \mathfrak{F}_{\frac{11}{2}} + 3\beta^2 \mathfrak{F}_{\frac{9}{2}} \right) \right],$$

with all $\mathfrak{F}_a = \mathfrak{F}_a(0)$. Using $\sigma \leq \lambda^2$ and the exact ratios of the $\mathfrak{F}_a(0)$, the bracket is at least $\lambda^4 \beta^6 \mathfrak{F}_{\frac{15}{2}}(0) (1 - 345\beta^{-2} - 9360\beta^{-4})$ in absolute value, and the last factor is at least 0.64 for $\mu \geq 5$.

With $1/I_2(\beta) \geq \sqrt{2\pi} \beta e^{-\beta}$ (Lemma 5.2(ii)) this gives

$$|\xi \hat{\varphi}'_B(\lambda)| \geq 0.64 \cdot \frac{(2\pi)^{19/2}}{2^{15/2} \Gamma(\frac{17}{2})} \lambda^{22} e^{-2\pi\lambda^2} \geq 9.6 \lambda^{22} e^{-2\pi\lambda^2} \quad (\mu \geq 5).$$

In the same way, now with leading term $\lambda^4 \beta^4 \mathfrak{F}_{\frac{13}{2}}(0)$ and factor $1 - 221\beta^{-2} - 1716\beta^{-4} \geq 0.77$, we get $|\hat{\varphi}_B(\lambda)| \geq 0.77 \cdot \frac{(2\pi)^{15/2}}{2^{13/2} \Gamma(\frac{15}{2})} \lambda^{18} e^{-2\pi\lambda^2} \geq 4.4 \lambda^{18} e^{-2\pi\lambda^2}$. Hence the powers 18 and 22 in (5.3) are optimal. This concerns Lemma 5.3 only. It does not show that $p = 46$ is optimal, since B_{φ_B} is an integral over $\xi \geq \lambda$, nor does it give any lower bound for $\lambda_{\min}$; a lower bound for $\lambda_{\min}$ would be of the strength of RH.

5.3. Gaussian limit.

Lemma 5.5. *For $|x| \leq \lambda$,*

$$\frac{\Phi_\beta(x/\lambda)}{I_2(\beta)} \leq C_{\text{dom}} e^{-\pi x^2}, \quad C_{\text{dom}} = \frac{\sqrt{2}}{\sqrt{2\pi} c_2} = 36.14 \dots$$

Moreover, as $\lambda \rightarrow \infty$, $\Phi_\beta(x/\lambda)/I_2(\beta) \rightarrow e^{-\pi x^2}$ uniformly on compact sets, $\sigma \rightarrow \frac{3}{2\pi}$, and $\varphi_B \rightarrow \pi^{-2}h$ uniformly and in $L^1((1+|x|)^N dx)$ for every N .

Proof. Put $t = x/\lambda$. From $\sqrt{1-t^2} \leq 1 - \frac{t^2}{2}$ we get $\beta \sqrt{1-t^2} \leq \beta - \pi x^2 =: \beta - d$, with $0 \leq d \leq \beta/2$. Since I_2 is increasing and $1-t^2 \leq 1$, Lemma 5.2(ii),(iii) give

$$\frac{\Phi_\beta(t)}{I_2(\beta)} \leq \frac{I_2(\beta-d)}{I_2(\beta)} \leq \frac{e^{\beta-d} / \sqrt{2\pi(\beta-d)}}{c_2 e^\beta / \sqrt{\beta}} \leq \frac{\sqrt{2}}{\sqrt{2\pi} c_2} e^{-d}.$$

For the limits, $\beta \sqrt{1-x^2/\lambda^2} = \beta - \pi x^2 + O(x^4/\lambda^2)$ and $I_2(\beta-d)/I_2(\beta) = e^{-d}(1 + O((1+d)/\beta))$ by the asymptotic expansion of I_2 . Dominated convergence, with the dominating function $C_{\text{dom}}(1+x^4)e^{-\pi x^2}$, gives $\sigma \rightarrow \int x^4 e^{-\pi x^2} / \int x^2 e^{-\pi x^2} = \frac{3}{2\pi}$ and the remaining claims, since $x^2(x^2 - \frac{3}{2\pi})e^{-\pi x^2} = \pi^{-2}h(x)$. $\square$

Numerically, $\sup I_2(\beta-d)e^d/I_2(\beta)$ over $\beta \in [2, 1000]$, $0 \leq d \leq \beta/2$, is 1.4116, so C_{dom} is far from sharp.

5.4. Conclusion. Let e_B and k_B be the functions e and k of Section 3 for $\varphi = \varphi_B$, and let e_h be as in Lemma 4.2.

Lemma 5.6. *There is λ_0 such that $\|k_B\|^2 \geq \frac{1}{2}\pi^{-4} \cdot 0.0093256$ for $\lambda \geq \lambda_0$.*

Proof. On $|y| \leq 0.1$,

$$|e_B(y) - \pi^{-2}e_h(y)| \leq e^{0.05} \sum_{n \geq 1} |\varphi_B - \pi^{-2}h|(ne^y).$$

The right-hand side tends to 0 uniformly in $|y| \leq 0.1$: for each fixed n by the uniform convergence in Lemma 5.5, and the tail $n \geq N$ is small uniformly in λ because $|\varphi_B| + \pi^{-2}|h| \leq C(1+x^4)e^{-\pi x^2}$ by Lemma 5.5, since σ is bounded for large λ . By Minkowski's inequality and $\int_{-0.1}^{0.1} e_h^2 \geq 0.0093256$ (proof of Lemma 4.2), the claim follows. $\square$

The threshold λ_0 is not effective: an explicit value would need a quantitative form of the convergence in Lemma 5.5.

By (5.3),

$$T_0(m\lambda) \leq \frac{2}{3}C\lambda^{19}e^{-2\pi\lambda^2}m^{-3/2}, \quad T_1(m\lambda) \leq 2C\lambda^{23}e^{-2\pi\lambda^2}m^{-1/2},$$

$$\lambda|\hat{\varphi}_B(m\lambda)| \leq C\lambda^{19}e^{-2\pi\lambda^2}m^{-5/2}.$$

Summing over m and using $\lambda \geq 1$ gives $B_{\varphi_B}(\lambda) \leq C'\lambda^{23}e^{-2\pi\lambda^2}$ with $C' = 2C(\zeta(\frac{5}{2}) + \zeta(\frac{3}{2}))$. Proposition 3.7 with Lemma 5.6 gives

$$\lambda_{\min}(\lambda) \leq \frac{2S C'^2 \lambda^{46} e^{-4\pi\lambda^2}}{\frac{1}{2}\pi^{-4} \cdot 0.0093256} \quad (\lambda \geq \lambda_0),$$

which is Theorem B.

As a numerical check, for the same φ_B the bound $2S(A_0 + A_1)^2$ of Proposition 3.5 exceeds $|W(k_B)|$ by factors 46, 85 and 207 at $\mu = 5, 7, 11$.

6. NUMERICAL EVIDENCE

6.1. The prolate quantity $1 - \chi_2$. We follow the notation of [6, §6]. The prolate spheroidal wave functions [14] for the window are $h_{n,\lambda}(x) = \text{PS}_{n,0}(2\pi\lambda^2, x/\lambda)$, supported in $[-\lambda, \lambda]$. The Fourier transform of $h_{2m,\lambda}$, restricted to $[-\lambda, \lambda]$, is $\chi_m h_{2m,\lambda}$, where $\chi_m^2 = \nu_m$ is the corresponding eigenvalue of time- and band-limiting and the sign of χ_m is $(-1)^m$. Thus χ_2 belongs to $h_{4,\lambda}$. By a theorem of Fuchs [9, Thm. 1], quoted in [4] and [6, footnote 19],

$$1 - \chi_2(\lambda) \sim \frac{2^{14}}{3} \sqrt{2} \pi^5 \mu^{9/2} e^{-4\pi\mu} \quad (\mu = \lambda^2 \rightarrow \infty).$$

In the tables we use this asymptotic formula. At $\mu = 11$ it agrees with our direct computation of the prolate eigenvalue to within 13%.

6.2. The bottom of the spectrum. Restricting to the even sector, we computed the bottom $\lambda_{\min}^{\text{even}}$ of W_λ with a Legendre–Galerkin discretization of dimension N ; the matrix entries are computed in ball arithmetic and the eigenvalues at high precision. The ground state is believed to be even; Zhu [18, Thm. 6.2] certifies this for $L = 0.8$ ($\mu = 4.95$). By the Rayleigh–Ritz principle, the Galerkin values are upper bounds for $\lambda_{\min}^{\text{even}}$, and they decrease as N grows. We also computed the Rayleigh quotient of the vector k_λ of [4], which is $\mathcal{E}(\varphi)$ restricted to the window for a combination φ of $h_{0,\lambda}$ and $h_{4,\lambda}$, projected onto the same basis.

μ	N	$\lambda_{\min}^{\text{even}}$ (Galerkin)	$W(k_\lambda)/\ k_\lambda\ ^2$	$1 - \chi_2$	$W(k_\lambda)/\ k_\lambda\ ^2(1 - \chi_2)$
5	140	9.27×10^{-18}	1.40×10^{-17}	1.70×10^{-18}	8.2
7	140	6.75×10^{-28}	9.33×10^{-28}	9.42×10^{-29}	9.9
13	260	2.54×10^{-59}	3.49×10^{-59}	2.75×10^{-60}	12.7
15	260	6.04×10^{-70}	8.72×10^{-70}	6.36×10^{-71}	13.7
17	300	1.35×10^{-80}	1.87×10^{-80}	1.36×10^{-81}	13.7
20	380	1.35×10^{-96}	1.80×10^{-96}	1.20×10^{-97}	15.0

Convergence in N was checked only partially. The change from the next smaller N that we computed was 5% at $\mu = 13$ ($N = 220 \rightarrow 260$), a factor 2.7 at $\mu = 15$ ($220 \rightarrow 260$), a factor 3.8 at $\mu = 17$ ($260 \rightarrow 300$) and 30% at $\mu = 20$ ($340 \rightarrow 380$); at $\mu = 5$ and 7 only $N = 140$ was computed. The values for $\mu \geq 15$ should therefore be read as upper bounds that may not have converged. At $\mu = 5$ our value is consistent with Zhu's certified enclosure $8.9 \times 10^{-18} \leq \lambda_1^{\text{even}} \leq 2.523 \times 10^{-16}$ for the slightly smaller window $\mu = e^{1.6}$ [18, Thm. 6.2]. The data are consistent with $\lambda_{\min} \asymp \mu^{9/2} e^{-4\pi\mu}$, as observed in [4, Fig. 4] and [6, Fig. 1].

6.3. Comparison with Zhu's certified bounds. Zhu [18, Table 3] gives certified upper bounds $\lambda_{\text{Zhu}}^*(L)$ for windows $[-L, L]$, so $\mu = e^{2L}$. Divided by $1 - \chi_2$, they stay within a narrow band over 283 orders of magnitude:

L	μ	λ_{Zhu}^* (upper bound)	$1 - \chi_2$	ratio
0.8	4.95	2.27×10^{-17}	2.95×10^{-18}	7.7
1.2	11.0	9.98×10^{-49}	8.03×10^{-50}	12.4
1.6	24.5	8.61×10^{-121}	5.51×10^{-122}	15.6
2.0	54.6	3.19×10^{-283}	1.66×10^{-284}	19.2

Over the whole range $0.5 \leq L \leq 2$ of [18, Table 3] the ratio lies between 3.4 and 19.2.

- The ratio grows slowly, like a small power of μ . This is compatible with $-\ln \lambda^* = 4\pi\mu - O(\log \mu)$.
- Zhu fits the empirical law $-\ln \lambda^* \simeq 2\pi^2 N(T^*)/\ln N(T^*)$, with a fitted constant. It behaves like $2\pi^2\mu$ as $\mu \rightarrow \infty$, but in this range it agrees numerically with $4\pi\mu$.
- Both sets of numbers are upper bounds. Deciding the true asymptotics requires lower bounds, which are of RH strength.

7. FORMALIZATION

The Lean statement `RhWeil.Concrete.theoremA` reads

$$\exists C p \lambda_0, \forall \lambda \geq \lambda_0, \exists k : \mathbb{R} \rightarrow \mathbb{C}, k \in C^2, \text{tsupport } k \subseteq [-\log \lambda, \log \lambda], 0 < \|k\|_2^2, \text{ and } \|\text{literatureRHS}(k \star \tilde{k})\| \leq C \lambda^p e^{-2\pi\lambda^2} \|k\|_2^2.$$

Here `literatureRHS` is the name used in [17] for the geometric side of Weil's explicit formula, in the standard form of [10, Thm. 5.12]:

$$\hat{g}(\tfrac{i}{2}) + \hat{g}(-\tfrac{i}{2}) - \sum_n \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) + \frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}(r) \left[\text{Re} \frac{\Gamma'}{\Gamma} \left(\tfrac{1}{4} + \tfrac{ir}{2} \right) - \log \pi \right] dr.$$

For $g = k \star \tilde{k}$ this is the Weil quadratic form $W(k) = QW(k, k)$. As a numerical check of the conventions, take $k(y) = (1 - y^2/a^2)_+^3$ and $\mu = 5$. The geometric side, the zero side (the first 1000 zeros) and a Legendre–Galerkin discretization all give $2.7877938966 \times 10^{-5}$, and they agree to about 10^{-12} .

The formal proof follows a route slightly different from Section 4:

- the window cut-off is smooth, $k = e\eta$, so that no jump occurs;
- Lemma 3.6 is replaced by a sup-type bound: if $|\hat{\varphi}| \leq M/\xi^2$ on $|\xi| \geq \Xi$, then $|\sum_n \varphi(nu)| \leq M\zeta(2)u$ for $u \leq 1/\Xi$;

- the constants are existential.

The radical lemma is proved through Mathlib’s Mellin transform and the identity theorem. Poisson summation is taken from Mathlib. The explicit formula is taken from [17]. The lower bound $\|k\|^2 \geq c_0$ uses no numerics: every summand $\varphi((n+1)e^y)$ with $n \geq 1$ is nonnegative. The development has about 3,300 lines. `#printaxioms` reports `propext`, `Classical.choice` and `Quot.sound`.

What is *not* formalized:

- the identification of the Rayleigh quotient with the spectral bottom [4, Thm. 3.6];
- the explicit constant 1.563×10^9 (proved on paper with ball arithmetic);
- Theorem B, which would require a theory of Bessel functions that Mathlib does not yet provide.

8. REMARKS

- (1) **What the theorems say.** Any proof of RH by window-by-window positivity must beat a margin that is, unconditionally, at most $e^{-4\pi\mu + O(\log \mu)}$.
- (2) **The polynomial factor.** It depends on the construction. The proof of Theorem B gives λ^{46} without optimization; numerically the bound of Proposition 3.7 for φ_B grows like $\mu^{16.8}e^{-4\pi\mu}$, against about $\mu^{4.5}e^{-4\pi\mu}$ for $W(k_\lambda)$.
- (3) **Relation to the CCM vector k_λ .** The vector k_λ is of the form $\mathcal{E}(\varphi)$ restricted to the window, with φ built from truncated prolate functions. The truncated prolate functions do not vanish at $\pm\lambda$, so φ does not satisfy (H), and we have not verified whether a variant of our argument bounds $W(k_\lambda)$ by a multiple of $1 - \chi_2$. We leave this as future work. Numerically $W(k_\lambda)/\|k_\lambda\|^2 \approx (8-15)(1 - \chi_2)$ for $5 \leq \mu \leq 20$.

REPRODUCIBILITY

The Lean development, the numerical scripts and their outputs, and a summary of the verification are available at <https://github.com/moriryota/weil-window-upper-bounds> and archived on Zenodo, doi:10.5281/zenodo.23059099 (all versions; this paper corresponds to release v1.0.1). The Lean development uses Lean v4.33.0-rc2, Mathlib revision 51e6992e, and the Zeta23 formalization at formal-math commit 3635e748. This paper is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0). The code is distributed under the Apache License 2.0.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude, Google Gemini and OpenAI GPT. Under the direction of the author, these systems did the following:

- drafted the proofs and the Lean development;
- implemented and ran the numerical computations;
- reviewed one another’s drafts adversarially;
- searched the literature;
- drafted this manuscript.

The author chose the problem and the strategy, assigned and reviewed the tasks, and takes full responsibility for the content. The correctness of Theorem A in its existential form does not rest on the AI systems, since it was checked by the Lean kernel. The explicit constants rest on the ball-arithmetic computations described above. Theorem B and the correspondence between the prose and the formal statement were checked by the AI systems and have not yet been reviewed by independent human experts. The AI systems are not authors of this paper.

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